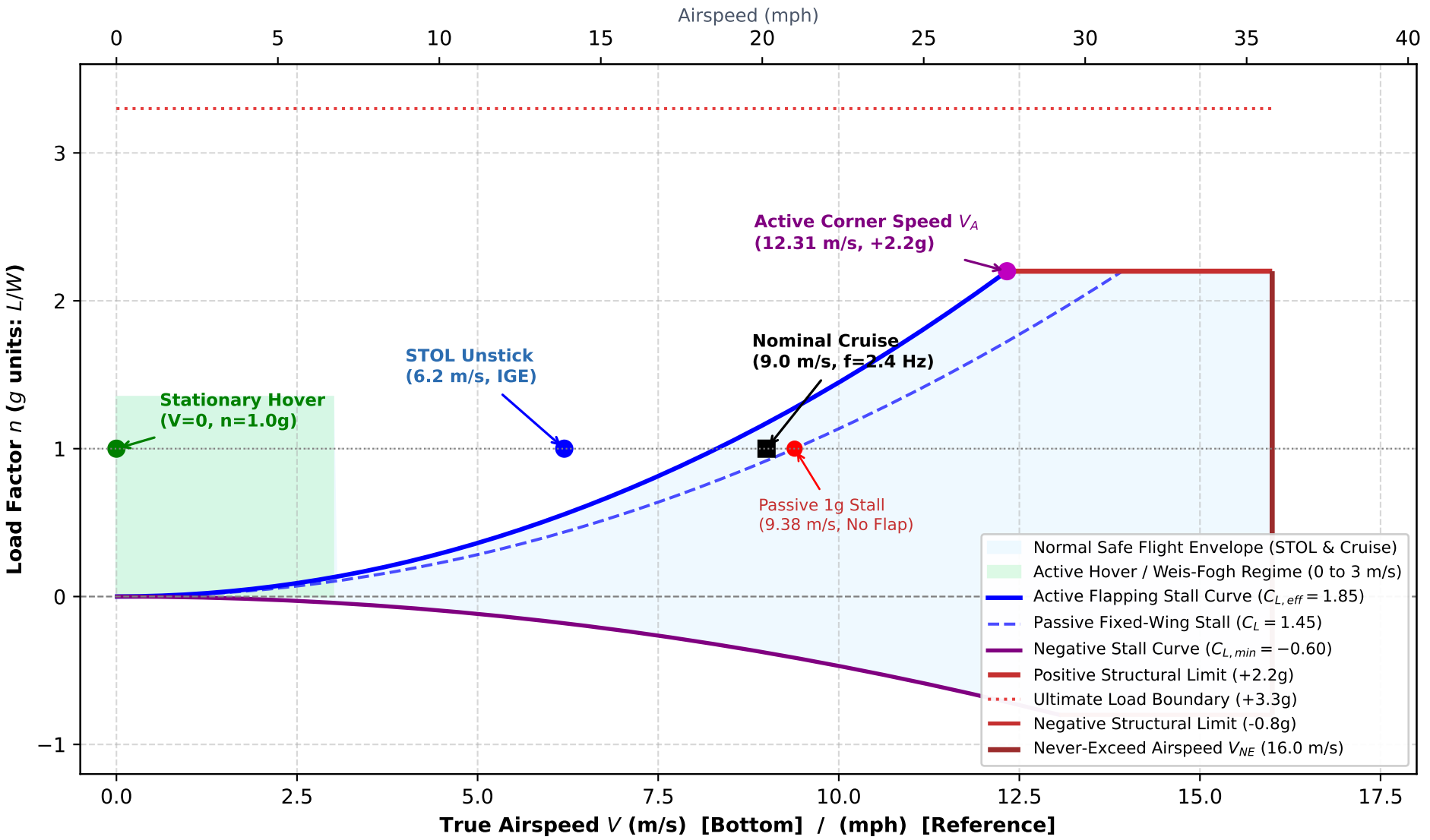
MCO-1D Mark 1 — V-n Structural Load Factor Flight Envelope
(MTOW = 5.687 kg, 8 Wings, S = 0.7124 m², 10:1 Capstan Reduction)
Airspeed (mph)
0
5
10
15
20
25
30
35
40
Load Factor n (g units: L/W)
3
2
1
0
−1
Active Corner Speed V_A
(12.31 m/s, +2.2g)
Nominal Cruise
(9.0 m/s, f=2.4 Hz)
STOL Unstick
(6.2 m/s, IGE)
Stationary Hover
(V=0, n=1.0g)
Passive 1g Stall
(9.38 m/s, No Flap)
Normal Safe Flight Envelope (STOL & Cruise)
Active Hover / Weis-Fogh Regime (0 to 3 m/s)
Active Flapping Stall Curve (C_L,eff = 1.85)
Passive Fixed-Wing Stall (C_L = 1.45)
Negative Stall Curve (C_L,min = −0.60)
Positive Structural Limit (+2.2g)
Ultimate Load Boundary (+3.3g)
Negative Structural Limit (-0.8g)
Never-Exceed Airspeed V_NE (16.0 m/s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
True Airspeed V (m/s) [Bottom] / (mph) [Reference]